Malhar Mangtani

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Portfolio: malharmangtani.github.io

Educational Information

Columbia University, New York City, USA
M.S. in Computer Science **Expected Dec 2027**
- Admitted for Fall 2026

Sarvajanik College of Engineering and Technology, Surat, India
B. Tech in Computer Engineering **Sep 2022– May 2026**
- Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Machine Vision, Python for Data Science, Data Structures & Algorithms, Database Management Systems, Operating Systems, Cloud Computing, Compiler Design.
- Cumulative Performance Index (CPI): **9.35/10.00 |** Degree CGPA: **9.11/10.00**

Sarvajanik College of Engineering and Technology, Surat, India
Honors in Artificial Intelligence/ Machine Learning **Dec 2023 – Dec 2025**
- Relevant Coursework: Introduction to Artificial Intelligence and Robotics, Machine Learning and Applications, Deep Learning, Modern trends in AI-ML.
- CGPA: **10.00/10.00**

Laurentian University, Sudbury, Canada
International Experience Program (IEP) **Jul–Aug 2024**
- Relevant Coursework: Mobile Computing and Wireless Communication, Machine Learning.

Work Experience

Oklahoma State University, Industrial Training and Assessment Center (ITAC), USA **Feb–Apr 2026**
AI Intern (Hybrid)
- Developed SmartITAC — an AI chatbot enabling the ITAC team to query 167,000+ energy assessment records through plain English, reducing a 1–2 hour task to seconds.
- Built an end-to-end RAG pipeline using HuggingFace embeddings, ChromaDB, and Groq LLM API with a multi-mode query system supporting keyword search, geographic filtering, statistical aggregation, and time-based filtering.
- Built a FastAPI backend with real-time streaming and an HTML chat frontend accessible to all team roles.
- Received official internship completion letter from Dr. Hitesh Vora, Oklahoma State University, funded by U.S. Department of Energy (DOE).

USICAMM, Secretaría de Educacíon Pública (SEP), Mexico **Jan–Jul 2025**
AI Intern (Remote)
- Contributed to an AI-driven teacher management platform for registration, promotion, and recognition.
- Built AI-based background check, profile evaluation, and validation search engine modules.
- Received official experience letter from USICAMM (SEP, Mexico) for significant contributions.

Laurentian University, Sudbury, Canada **Jul–Aug 2024**
AI/ ML Research Intern (On-site)
- Built a DeepLabV3 + ResNet50 model for nuclei segmentation and predictive analysis in histopathology images.
- Achieved high precision, recall, and F1-scores, supporting reliable diagnostic workflows.
- **Tools:** Python, TensorFlow, DeepLabV3, ResNet50

Research Projects and Presentations

Synthetic CT Generation from MRI **2025**
- Built and iteratively improved a deep learning pipeline for MRI-to-CT synthesis, progressing from a 3D Pix2Pix GAN (3D U-Net generator) to an anatomy-aware SwinGAN (SwinUNETR generator), both paired with a ConvNeXt-based anatomical classifier.
- Achieved stronger results with SwinGAN across all anatomical regions (avg. MS-SSIM **0.932**, PSNR **27.83** dB, MAE **0.027**), with peak performance on Thorax (MS-SSIM **0.947**, PSNR **28.30** dB, MAE **0.026**), outperforming the Pix2Pix baseline with high structural fidelity in complex bone-air interface regions.

Malaria Parasite Detection **2025**

- Developed a malaria parasite detection system in thin blood smear images using ConvNeXt with Grad-CAM++ for explainable malaria detection.
- Resulted in an IEEE conference publication and was presented at ICETCI 2025, Hyderabad, India.
- Received the Best Presenter Award in my session.

Emotion Detection Using Deep Learning Algorithms **2025**

- Developed and evaluated multiple CNN architectures (ResNet50, EfficientNetB3, VGG19) for a 6-class image classification task.
- Achieved highest performance with ResNet50, reaching **93.36%** validation accuracy and **67.22%** test accuracy, supported by confusion matrices and detailed classification reports.

American Sign Language Recognition System **2024**

- Developed a real-time sign language recognition system using LSTMs with MediaPipe and OpenCV for hand and gesture tracking.
- Enabled accurate gesture-to-text translation to improve communication accessibility.
- Achieved **99%** training accuracy, with testing accuracy of **96.49%** in letter recognition, **99.995%** in number recognition, and **98.396%** in static word recognition, obtaining an average of **98.29%** based on gesture recognition.

Real-Time Face and Hand Recognition with Finger Count **2024**

- Implemented a recognition system integrating facial feature detection and hand gesture recognition.
- Developed a finger-counting module for real-time human–computer interaction.

Publications

Malaria Parasite Detection in Thin Blood Smear Images Using ConvNeXt with Explanation **2025**

- Published in the Proceedings of the IEEE International Conference on Emerging Techniques in Computational Intelligence (ICETCI 2025), Hyderabad, India. DOI: 10.1109/ICETCI67340.2025.11258060

Technical Skills and Interests

- **Programming Languages:** Python, Java
- **Natural Language Processing & Generative AI:** Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Prompt Engineering, HuggingFace Transformers, Sentence Embeddings, Vector Databases (ChromaDB), Groq API, LangChain
- **Computer Vision and Medical Imaging:** OpenCV, MONAI, SimpleITK/ITK, Image Processing, Image Augmentation, MRI/CT Analysis
- **Tools and Frameworks:** FastAPI, MediaPipe, Jupyter Notebook, Conda / Virtual Environments, SQL, VS Code
- **Specialized Models:** GANs, CNNs, RNNs, LSTMs, Transfer Learning, Transformer Models
- **Machine Learning and Deep Learning:** TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Hyperparameter Tuning (GridSearchCV, RandomizedSearchCV), Model Evaluation Metrics (Precision, Recall, F1-score, ROC-AUC), Kaggle
- **Areas of Interest:** Artificial Intelligence and Machine Learning, Deep Learning & Computer Vision, Natural Language Processing & Conversational AI, Retrieval-Augmented Generation, Data Science & Predictive Analytics, Explainable and Trustworthy AI, Generative AI (Diffusion Models, GANs, VAEs), Medical Imaging and Healthcare AI